

Project Update 4

ECON 345

Due: 11:59PM October 10, 2026

Instructions (Do Not Skip)

Completing a thoughtful research project doesn't happen overnight. Accordingly, we have designed 6 project update assignments that will guide you through the research process. These project updates build upon each other, meaning that you will need to complete Project Update #4 in order to successfully accomplish Project Update #5. Once you submit a project update, your project buddy and economist adviser will provide feedback on your work. If you thoughtfully complete all 6 project updates and routinely incorporate feedback from your buddy and adviser, you will be on track for a solid final research presentation and paper.

Project Update #4 forks depending on your data. You will:

- Incorporate weights into your analysis, **if your project has weights**, or
- Incorporate feedback from your adviser and buddy so far, **if your project does not have weights**

You will not receive credit for work that is missing or incomplete without an explanation of what you tried and how you're struggling. For any parts you are unable to run in RStudio and produce the output for, please include the code (even if it does not work) and briefly explain what you did to attempt the task, and what problems arose (were there specific error messages?). In order to receive partial credit, you must include the code you attempted.

If you run into any difficulties while trying to complete this assignment, consider the following resources:

- Email your adviser (w. econ345.spelman@gmail.com cc'd) with questions on research design. They want to hear from you often!
- For technical help, post a question on Canvas, email econ345.spelman@gmail.com, or come to office hours. Atlanta Fed staff are eager to help!

Instructions for Completing This Assignment (Do Not Skip)

For this assignment, you will need to submit i) your RMarkdown (RMD) file (download and use the `project_hw4_student.Rmd` template) and ii) the resulting PDF or Word document.

RMD File

The RMD file is where you will compose your project's production code alongside written interpretation in order to easily and clearly communicate your research. As with earlier updates, organize pieces of your code into well-labeled chunks throughout your RMD file, then write your response/interpretation either above or below each chunk.

Ideally, you should knit your RMD to PDF so we can easily view your results and provide feedback. Alternatively, you can knit to Word. **If you cannot knit, submit screenshots of your output and your interpretation in a word document.**

If your project has weights, do Parts A, C, and D only. If your project does not have weights, do Parts B, C, and D only.

Part A - Incorporating Weights (80 points)

In order to have more accurate estimates, your analysis must incorporate weights, which you will do here. Consider using the `vtable` package for polished tables — see `summary_tables_with_vtable_package.pdf`, distributed with Lecture 4's materials, for an example.

1. For your summary statistics (30 points):

- Compute your analysis with weights (15)
- Interpret your results. Compare your results with your previous analysis. Are there significant changes? What does this tell you about your data? For example, is a certain group underrepresented in your sample before using weights? (15)

```
# compute weighted summary statistics
```

2. Histograms (30 points):

- Compute your analysis with weights (15)
- Interpret your results. Compare your results with your previous analysis — are there significant changes? (15)

```
# compute weighted histograms
```

3. Conditional summary statistics (20 points):

- Reproduce your charts from last homework with weights, if applicable, or create new ones. If not applicable, explain why. (10)
- Interpret your results — are there significant changes? (10)

```
# compute weighted conditional summary statistics
```

4. Conditional histograms (20 points):

- Compute your analysis with weights (10)
- Interpret your results — are there significant changes? (10)

```
# compute weighted conditional histograms
```

Part B - Incorporating Feedback (80 points)

1. State all the feedback from your adviser and buddy that you have received up to this point. (30 points)
2. Incorporate the feedback you have received from your adviser and buddy. For example, this may require you to reproduce your results or refine your writing. (30 points)

```
# reproduce or compute results if necessary
```

3. Take previous visuals, statistics, or charts and polish them — give your charts appropriate titles or captions, or use `kable()/vtable` to polish your tables. (20 points)

```
# polish previous analysis
```

Part C - Follow-up With Your Adviser (10 points)

1. Email your adviser with the knitted output (i.e., the PDF or Word document that results from your RMD). Do not send them your R script file or your RMD, but do submit these on Canvas. In your email, give your adviser a brief summary of what you've done this week and what you've learned about your research question, and include any questions or ideas about where to take your analysis next.
 - Email your knitted output to your adviser with econ345.spelman@gmail.com cc'd.

- Continue the conversation with your adviser by summarizing your findings and brainstorming next steps in the body of your email.

To receive credit for this, you must cc econ345.spelman@gmail.com.

Part D - Share With Your Classmates (10 points)

1. Post on the Canvas discussion board for the project a post that includes:

- Your research topic and a brief description of your data (5 points)
- A description of some of the challenges you've encountered so far (5 points)

You're strongly encouraged to comment on your classmates' posts as well!